

Can teleseismic data resolve a finite-fault model of an M_w 6.1?

The 2023 Mariana Islands earthquake (us6000kzv3) as a resolution test of the
MTUQ+WISP pipeline

Automated pipeline (17.Venezuela_2026)

2026-07-29

Purpose

USGS publishes finite-fault models routinely for $M_w \gtrsim 6.5$; for the 2023 M_w 6.1 Mariana Islands earthquake no finite-fault product exists. Hartzell, Mendoza & Zeng (2013, GRL) nevertheless resolved the M_w 5.8 Mineral, Virginia rupture from teleseismic P waves alone — their enabling ingredients being P waves to ~ 1 Hz (attenuation $t^* \approx 1$ s still passes source detail at teleseismic range), resolution carried by timing rather than amplitude, km-scale subfaults, and a compact fault. This report applies that recipe with the MTUQ+WISP pipeline and states explicitly what is, and is not, resolved.

1 Event and reference solution

Origin	2023-08-14T13:51:53.8 UTC, 13.347°N, 147.530°E, depth 8 km
Magnitude	M_w 6.1 (W-phase); $M_0 = 1.74 \times 10^{18}$ N m; centroid 13.5 km; DC 91 %
Tectonics	oblique normal + strike-slip east of Guam, near the trench (outer-rise style)
W-phase NP1	253.3/56.1/−28.7
W-phase NP2	000.3/66.5/−142.6
USGS products	MT, ShakeMap, PAGER, DYFI — no finite fault

2 Data

205 three-component stations, 32–89°, maximum azimuthal gap 28° — the densest coverage of any event in this project, which matters doubly here: the mechanism has a large strike-slip component, and the Kumamoto blind test showed clustered azimuths alias strike-slip radiation. MTUQ used an azimuth-balanced 60-station response-corrected subset; WISP applied its own quality control. WISP’s body-wave filter is 0.01–1.0 Hz — the Hartzell band — with auto-sized 2.4×2.0 km subfaults (255 total).

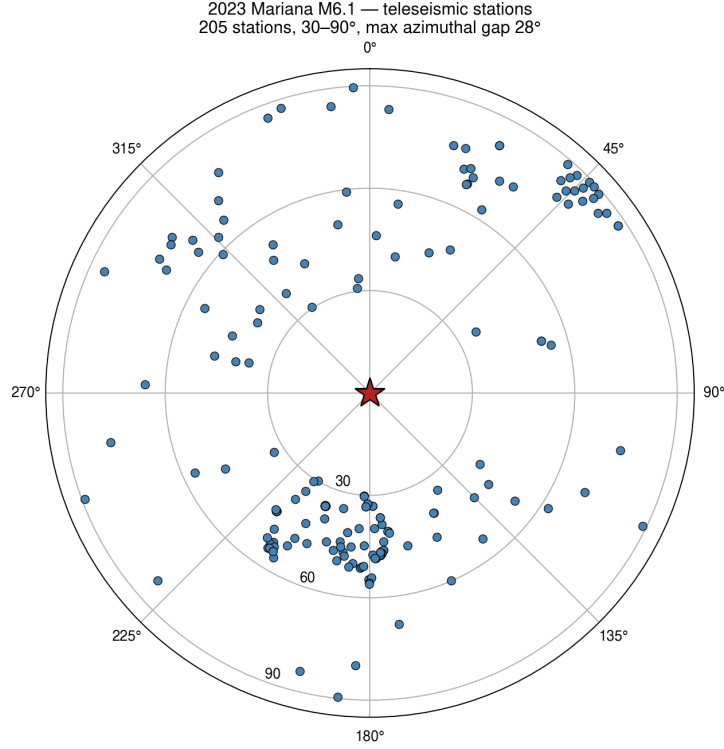


Figure 1: Teleseismic stations (azimuthal-equidistant; rings 30/60/90°).

3 Moment tensor (MTUQ): the independent point source fails at M_w 6.1

At $M_0 \sim 2 \times 10^{18}$ N m the standard teleseismic bands (body 10–33 s, surface 50–100 s) are low-SNR at 30–90°, and it shows:

variant	mechanism	M_w	Kagan vs W-phase
DC joint, 60 stations	68/82/−2	6.25	52.5°
DC body P+SH, 60 stations	122/54/52	5.80 (lower grid edge)	85.7°
depth scan 6–33 km	mechanism drifts	—	misfit varies 0.2 %

Table 1: Neither the mechanism nor the centroid depth is independently resolvable; the depth-misfit curve is flat to 0.2 % over 6–33 km. This is not an azimuth-coverage problem (gap 28°) — it is a signal-to-noise floor. Pipeline rule for $M_w \lesssim 6.5$: take mechanism and depth from the W-phase and spend the teleseismic data on the finite fault — exactly the Hartzell strategy.

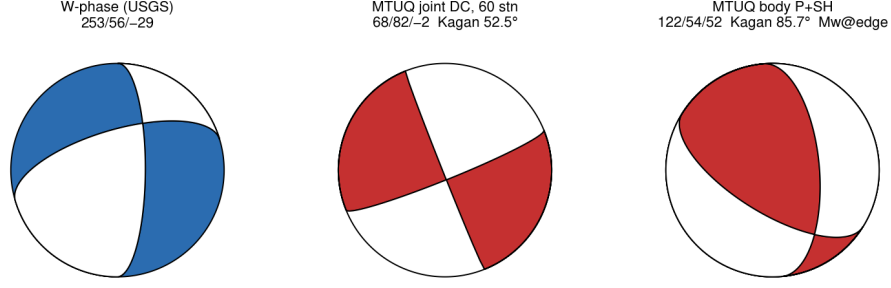


Figure 2: W-phase reference vs the two MTUQ attempts.

4 Finite fault (WISP): both W-phase planes, Mww-seeded, **shift_match** applied

plane	strike/dip/rake	misfit (post-shift)	peak slip	T_{5-95}
NP1 (preferred)	253/56/-29	0.2012	24.9 cm	8.2 s
NP2	0/66/-143	0.2058	24.9 cm	7.9 s

Table 2: Recovered $M_0 = 1.76 \times 10^{18}$ Nm vs W-phase 1.74×10^{18} — 1 % agreement. The 2.2 % NP1 preference sits between the Calama near-tie (1.6 %, ~ 10 km rupture) and the SPdA decisive margin (10 %, ~ 50 km rupture): suggestive, not conclusive, exactly as the rupture-size criterion predicts.

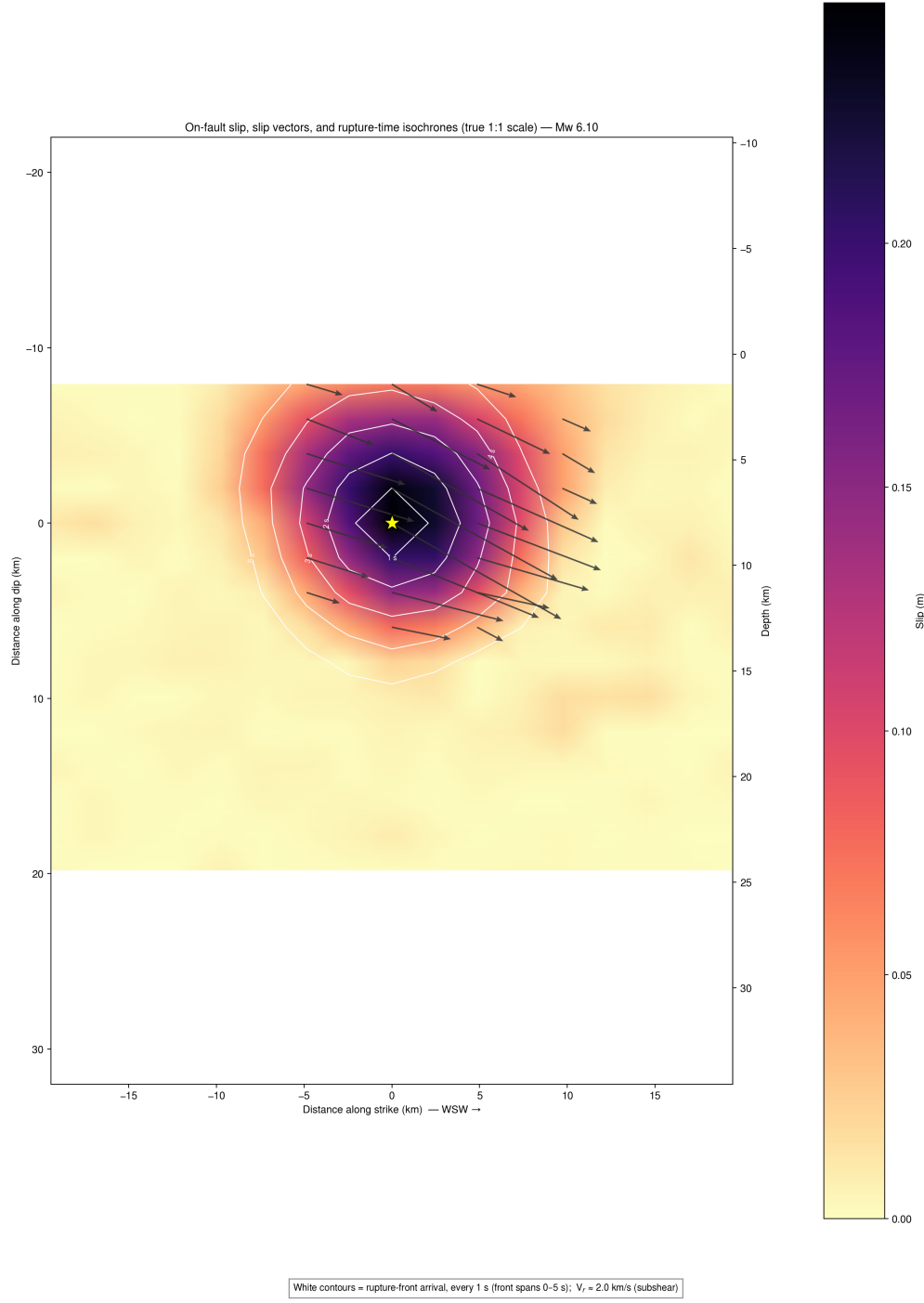


Figure 3: Preferred plane (253/56/−29), true 1:1 scale: a single compact asperity ~ 10 km across at 2–13 km depth, centred on the 8-km hypocentre; peak slip 25 cm; concentric 1-s isochrones spanning 0–5 s ($V_r \approx 2.0$ km s $^{-1}$, subshear); oblique right-lateral+normal slip vectors.

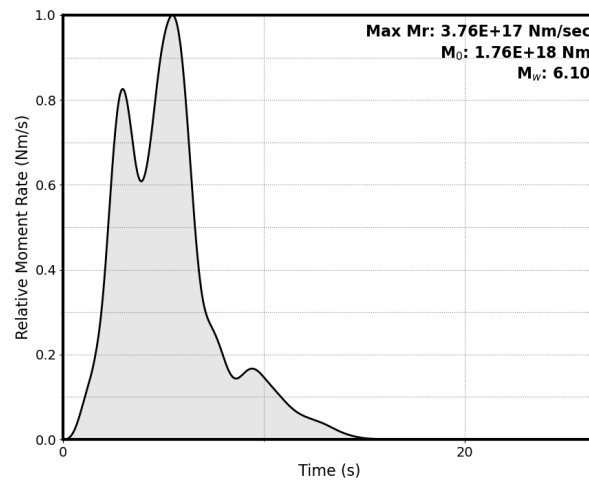


Figure 4: Moment-rate function (preferred plane).

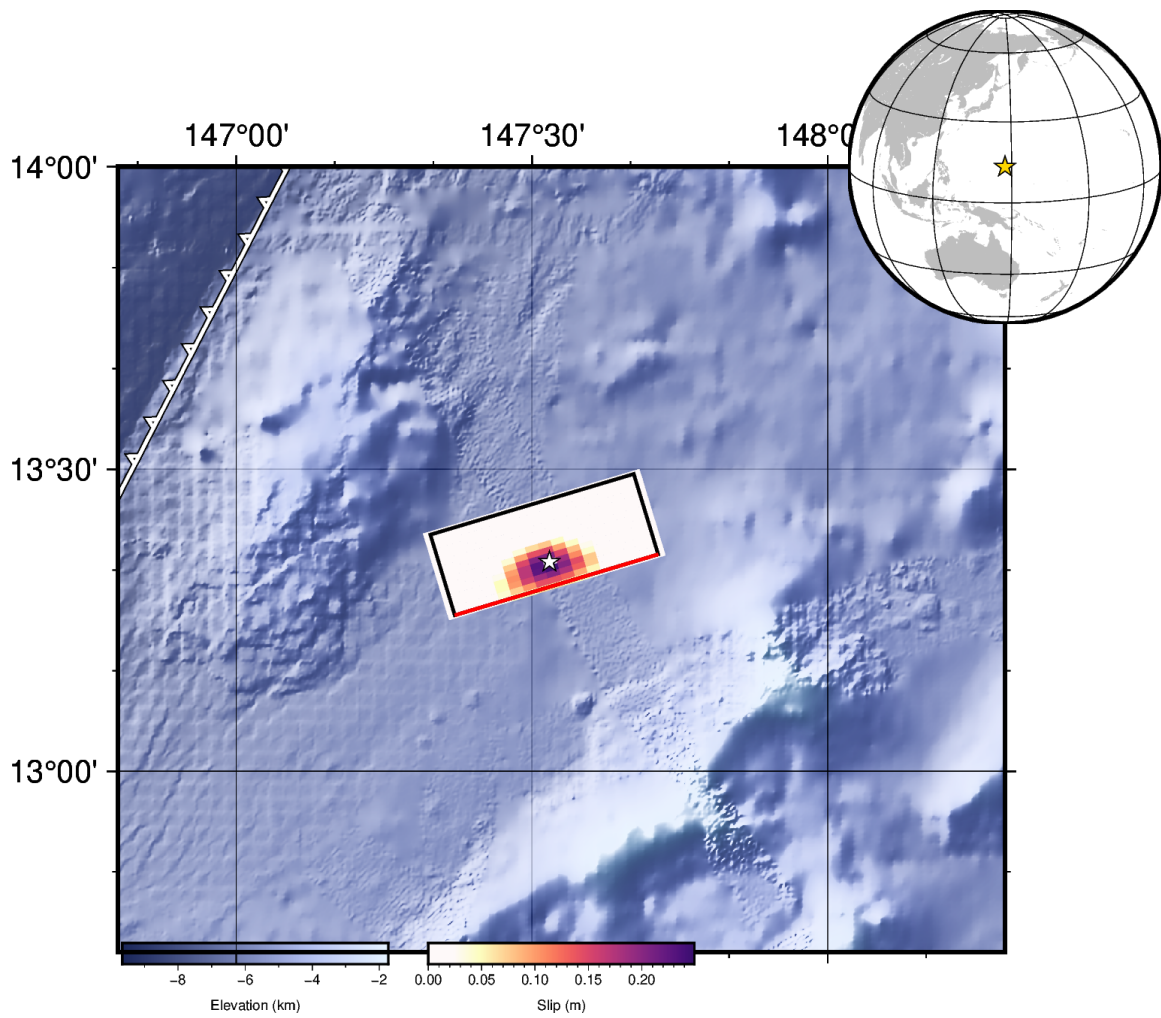


Figure 5: Map view.

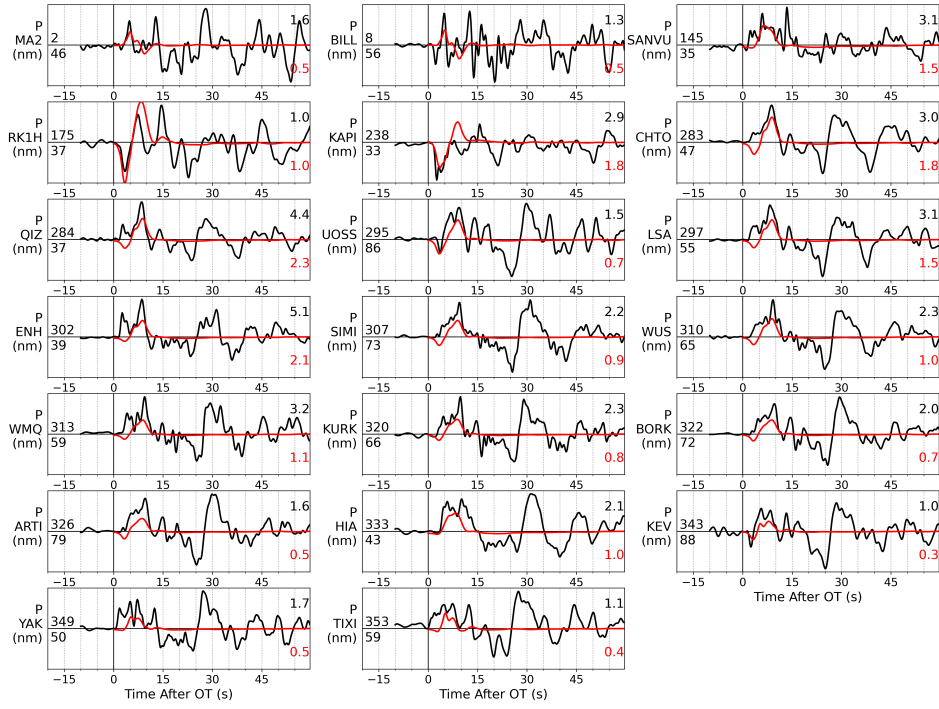


Figure 6: Teleseismic P fits (preferred plane, shifted) — the dataset that carries the Hartzell-style timing resolution.

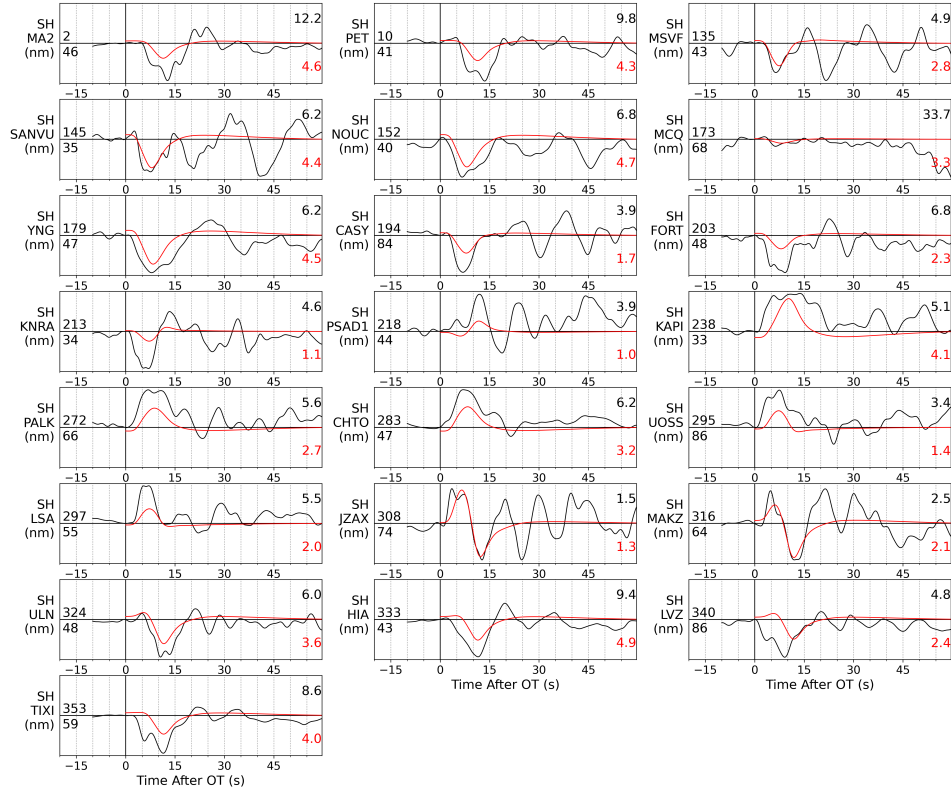


Figure 7: Teleseismic SH fits (preferred plane, shifted).

5 Resolution assessment

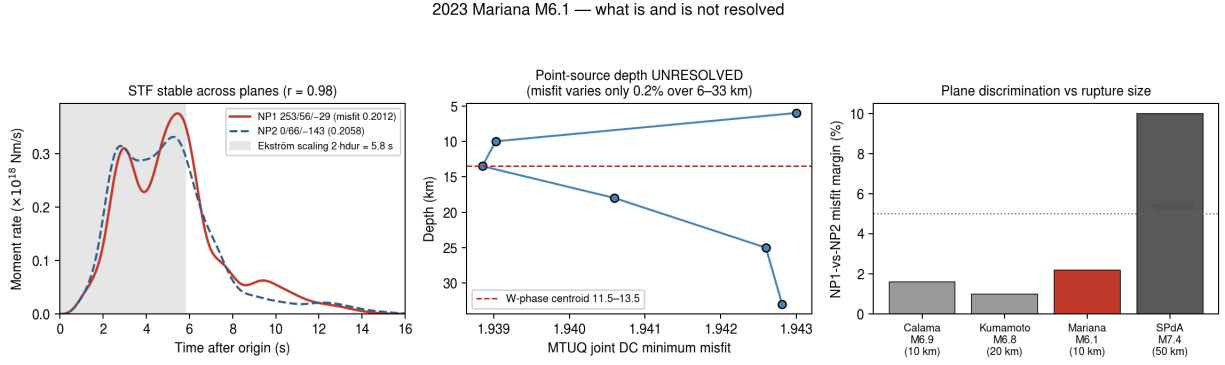


Figure 8: Left: the source-time function is stable across the two (very different) assumed planes, $r = 0.98$, including its two-pulse structure at 2.8 and 5.4 s — the plane-independent, therefore data-driven, part of the model. The Ekström-scaling duration band ($2 \cdot h_{dur} = 5.8$ s) captures the STF body; the tail beyond it is regularization. Centre: the MTUQ point-source depth scan is flat to 0.2% — depth must come from the W-phase. Right: the NP1-vs-NP2 misfit margin in the context of the pipeline’s plane-discrimination criterion.

Resolved (stable across planes, passing physical checks):

- a single compact asperity ~ 10 km across at 2–13 km depth, centred on the hypocentre — consistent with M_w 6.1 source scaling;
- the STF shape including its two-pulse structure (2.8 + 5.4 s), cross-plane $r = 0.98$;
- M_0 to 1% of the W-phase; rupture front spanning 0–5 s, $V_r \approx 2.0$ km s $^{-1}$;
- stress drop 1.0 MPa (effective-area convention, $(\Sigma u)^2 / \Sigma u^2 \rightarrow 263$ km 2) to 3.7 MPa (half-peak area, 110 km 2) — in the physically expected 1–10 MPa range.

Not resolved (and stated so):

- the fault plane: the 2.2% NP1 margin is suggestive, not decisive, at 10-km rupture size;
- an independent point-source mechanism or centroid depth (Section 3);
- fine slip detail and the duration tail: $T_{5-95} = 8.2$ s exceeds the 5.8-s scaling expectation because smoothing spreads low-level slip; the 25-cm peak is a lower bound;
- absolute updip/downdip placement within the compact patch.

6 Verdict

Yes: a reliable first-order finite-fault model of an M_w 6.1 is recoverable from teleseismic data — when the Hartzell recipe is followed (W-phase-fixed mechanism, P waves to 1 Hz, km-scale subfaults,

claims restricted to plane-stable features). What cannot be claimed at this magnitude: the fault plane, an independent mechanism, or a centroid depth from teleseismic point-source inversion. This extends the pipeline’s validated range from M_w 6.8 down to M_w 6.1, one magnitude unit above the Hartzell et al. (2013) M_w 5.8 benchmark.